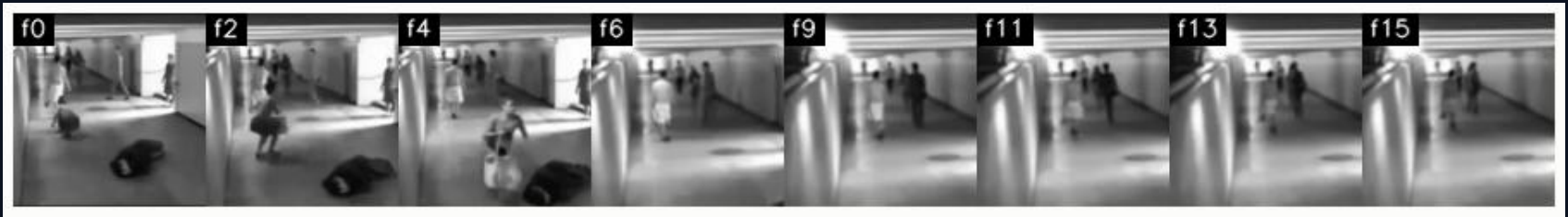
CSE4076 · IMAGE AND VIDEO ANALYTICS

Real-Time Illicit Activity Detection

What happens to the video before deep learning: processing and analytics, stage by stage



Sample clip produced by the pipeline (8 of 16 frames, 224 × 224 model input)

A CCTV system that flags illicit activity



Data preparation for these models is today's topic

0.907

mAP50 of the fine-tuned YOLOv11s (LLVIP + ExDARK)

70.2 ± 5.6 %

macro-F1 of VideoMAE, official 4-fold UCF-Crime split

4 classes

Fighting, Robbery, Vandalism, Normal (300 videos)

The old preprocessing had five silent gaps

Frame step = round(30 / 12) = 2	Clips run at 15 FPS, not 12: motion speed and the 5.3 s window are wrong
cv2.resize(frame, (640, 640))	4:3 frames get 33 % horizontal stretch: distorted bodies
Three brightness checks (50 / 80 / HSV)	The same frame is "dark" in one file and "bright" in another
16 uniform clips per video	Clips labelled Fighting often show no fighting: label noise
Validity check: clip.std() > 0.01	Frozen, blurry, black or cut-spanning clips all pass

Eight stages from raw video to clips

S1
Profile
PyAV stats, thresholds

S2
Resample
exactly 12 FPS

S3
Clean
cuts, bars, overlay, frozen

S4
Enhance
Zero-DCE++, denoise, deflicker

S5
Camera comp.
skipped: 2 / 300 videos move

S6
Motion
MOG2, optical flow, MHI

S7
Actors
YOLO, ByteTrack, crop

S8
Select
activity, proposals, gate

S9
Package
uint8 clips, sidecars

No model is trained here. Zero-DCE++ and YOLO weights are the existing ones; RAFT-small, FastDVDnet and TransNetV2 are public pretrained models; everything else is classical (OpenCV, NumPy, SciPy).

S1: profiling sets the thresholds

300

videos, all 30 FPS at 320 × 240

206

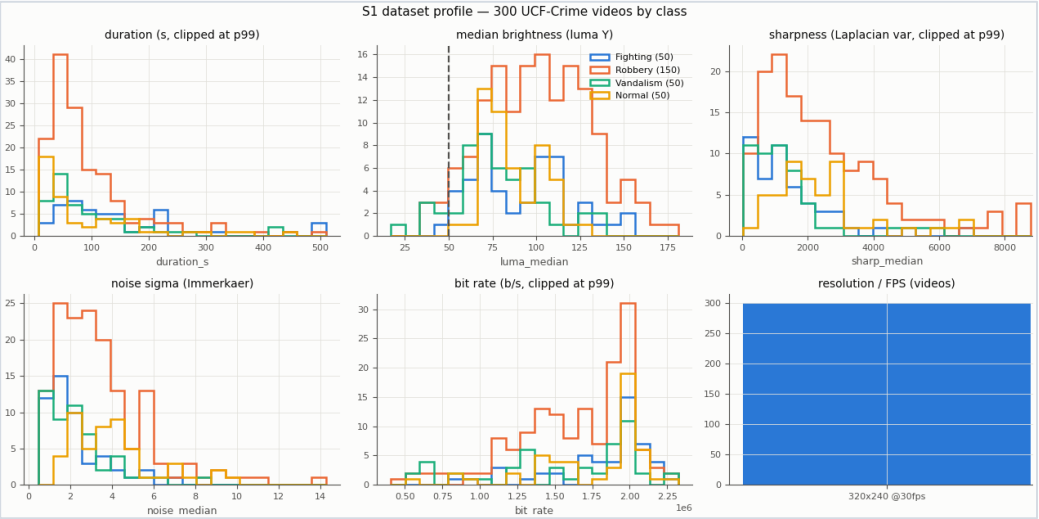
have black letterbox bars

218

carry a timestamp or logo overlay

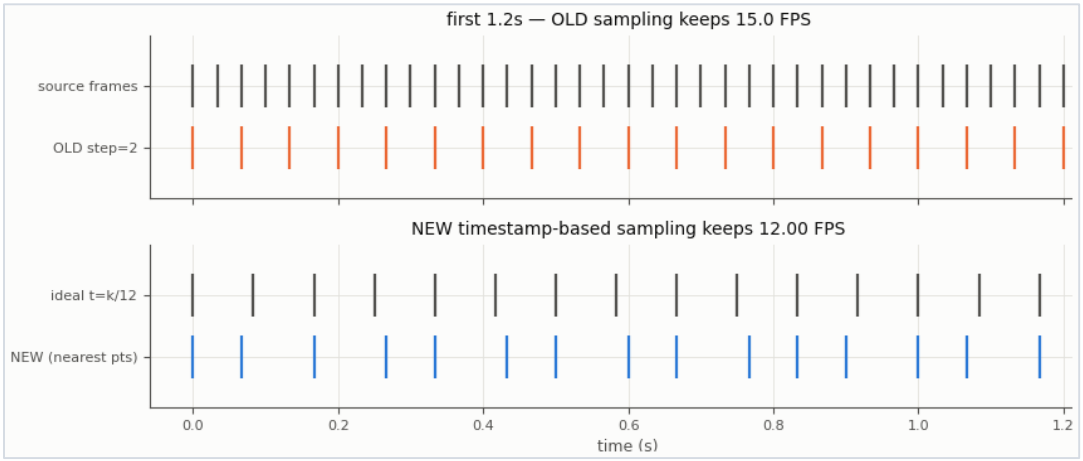
50

dark threshold: luma of the darkest 5 % of frames



Per-class histograms of all 300 videos (dashed line = dark threshold)

S2: exact 12 FPS from timestamps



Old rule keeps every 2nd frame (15 FPS); new rule picks the frame nearest to t = k/12 s

15.0 → 12.00

effective FPS of every clip

- PyAV decoding, max timing error 16.7 ms
- VideoMAE stride 4 now spans the intended 5.3 s
- Found on the way: many videos are only 10–15 unique FPS in a 30-FPS file, so motion is measured against the last distinct frame

S3: cleaning removes non-scene content



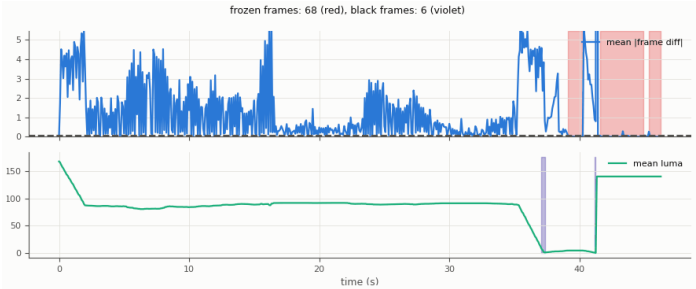
Letterbox bars cropped (206 / 300 videos) (input | output)



Timestamp and caption detected, inpainted (input | output)

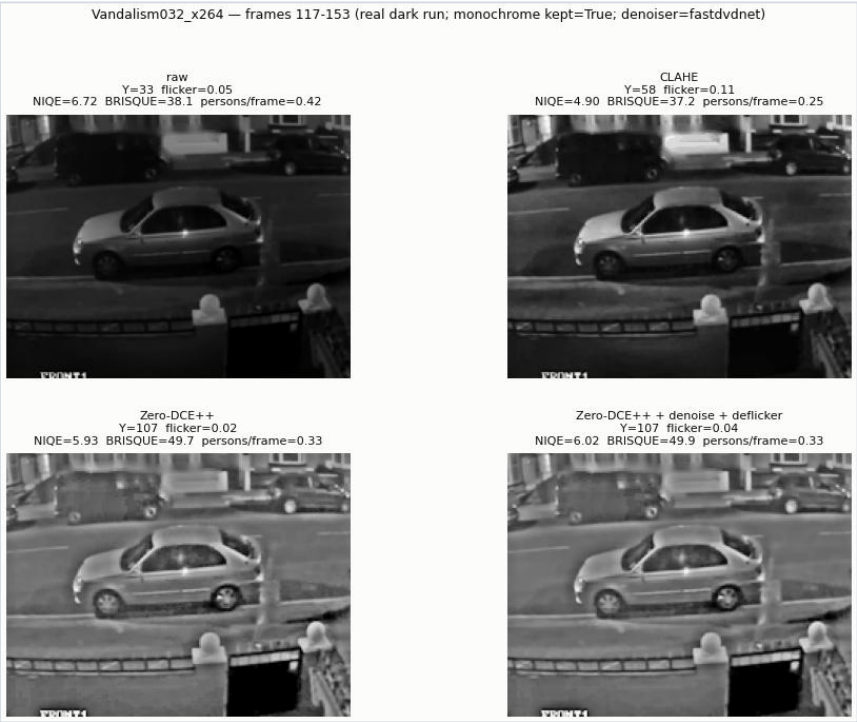


Shot cut found by HSV + edges and TransNetV2 (input | output)



Frozen and black frames flagged (68 in one video)

S4: enhance dark footage, then denoise



8 / 8

noise lowered by temporal denoising

7 / 8

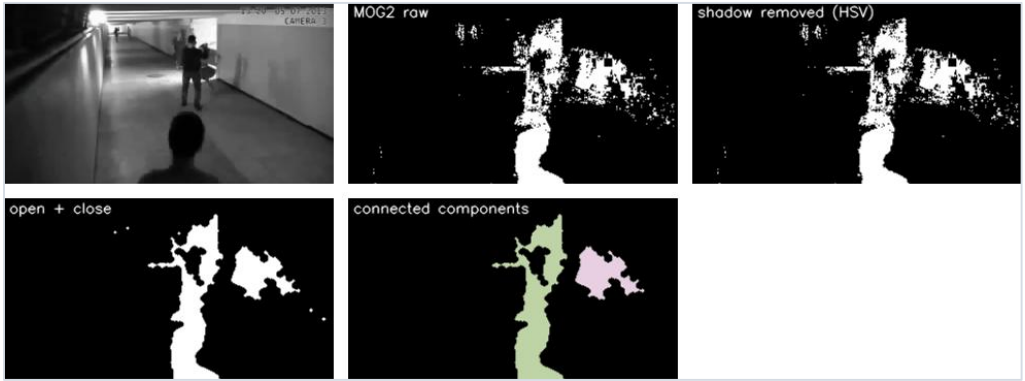
NIQE better than Zero-DCE++ alone

5 / 8

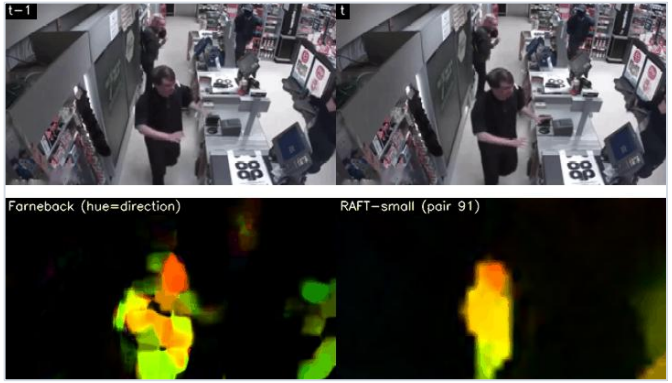
flicker index improved (already tiny in static scenes)

Real dark video shown; 7 of 8 demos use clearly labelled simulated low light.

S6: motion analysis turns video into signals



Input frame, MOG2 mask, shadows removed, open + close, connected components



Frame pair, then Farneback and RAFT-small flow (HSV colour)

- Foreground ratio and blob count; flow magnitude, direction entropy, divergence, curl
- Motion History Image: the last second of movement in one picture
- One motion-energy value per frame feeds clip selection (S8)

Finding: fine-tuned YOLO does not transfer



Top: fine-tuned best.pt (green). Bottom: COCO yolo11s.pt (orange). Same frames.

34 %

of fine-tuned boxes cover over a third of the frame

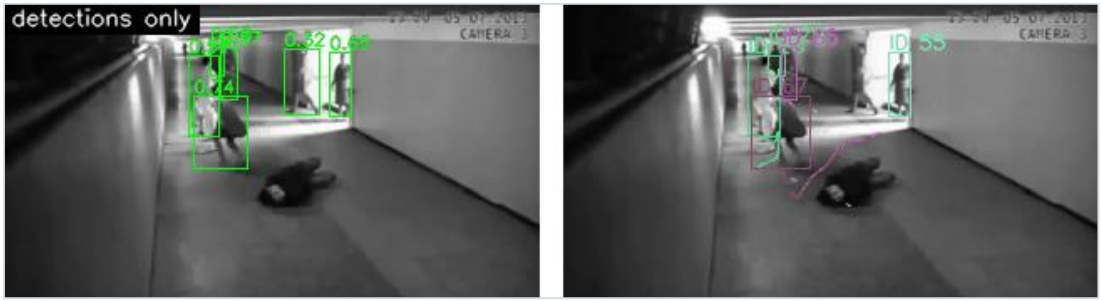
14 %

of its boxes are also found by the COCO model

COCO

yolo11s.pt chosen for S7; mAP 0.907 holds on LLVIP only

S7: tracks and an actor-centred crop



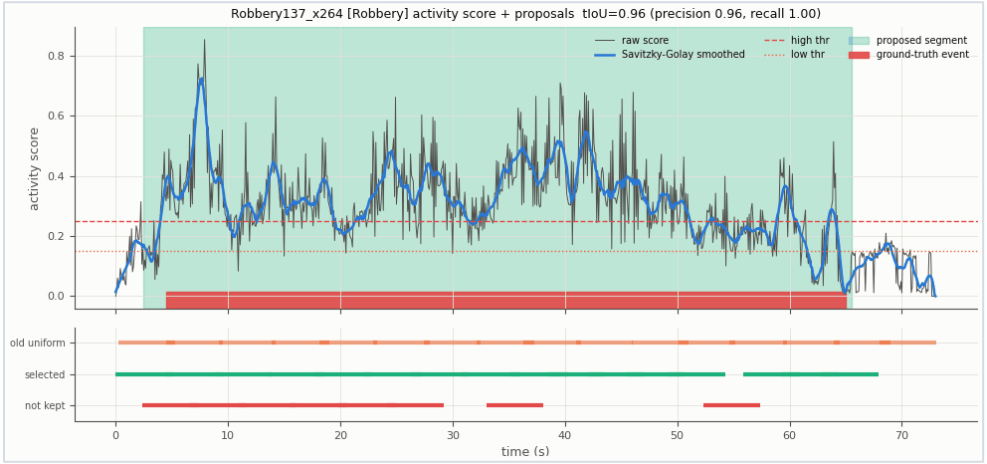
ByteTrack: persistent IDs and trajectories



Crop window (yellow) and the 224 × 224 model input

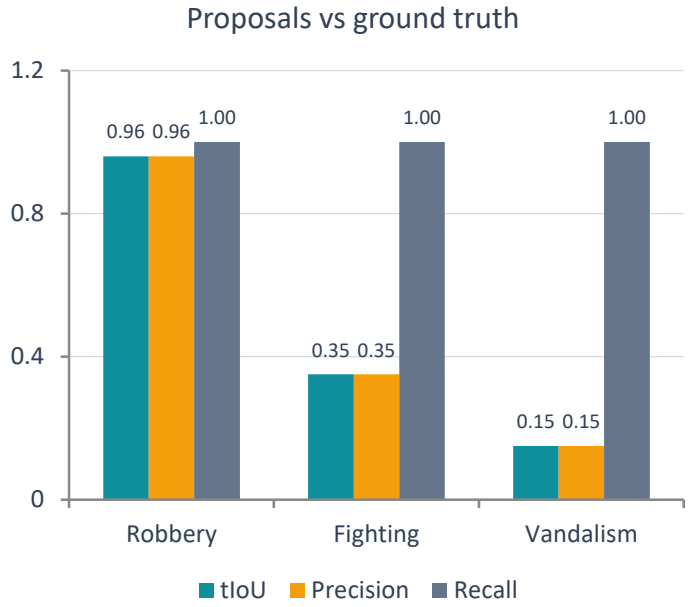
- Interaction signals: people count, minimum distance in person heights, box overlap, speed
- Crop: union of interacting people + 25 % margin, smoothed, letterboxed, INTER_AREA
- Crop jitter drops about 3×, e.g. 3.4 → 1.2 px

S8: sample clips where something happens



Activity score, proposed segments (green) and UCF-Crime ground truth (red bar)

Abnormal classes: clips inside proposals. Normal: uniform + 25 % hard negatives. Never across a shot cut.



Recall 1.0 everywhere; precision is limited because motion is not the same as crime.

S8: a quality gate that sees real defects



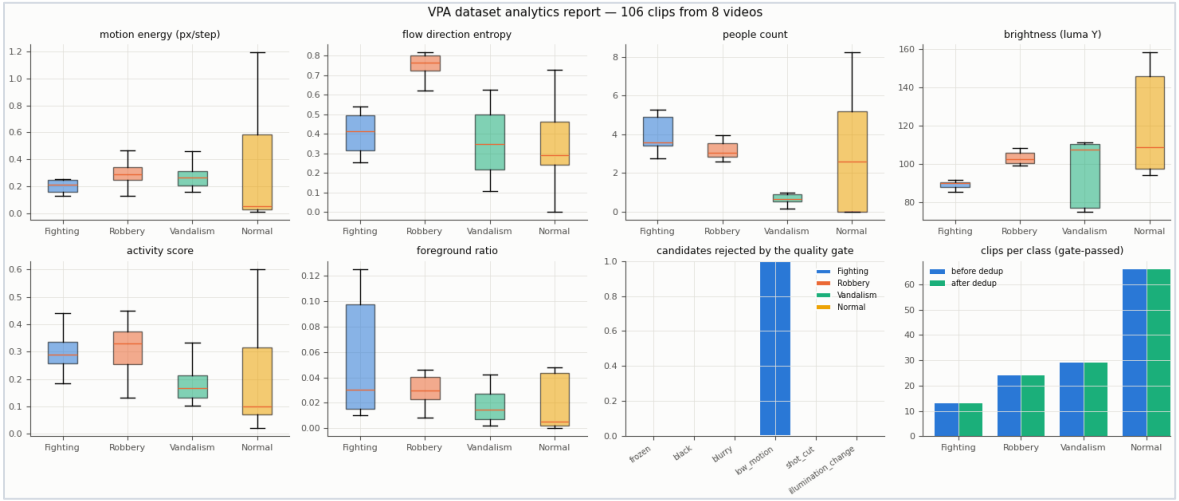
Fighting018_x264: same 16 uniform clips, old vs new check

#	start	OLD std>0.01	NEW quality gate
0	0.2s	pass	FAIL: shot_cut
1	2.8s	pass	pass
2	5.8s	pass	pass
3	8.9s	pass	pass
4	11.5s	pass	pass
5	14.5s	pass	pass
6	17.4s	pass	FAIL: low_motion
7	20.5s	pass	pass
8	23.0s	pass	pass
9	25.9s	pass	pass
10	29.2s	pass	FAIL: low_motion
11	31.8s	pass	FAIL: shot_cut
12	34.7s	pass	FAIL: shot_cut
13	37.7s	pass	FAIL: shot_cut
14	40.5s	pass	FAIL: frozen,shot_cut
15	41.2s	pass	FAIL: frozen

The same clips, two checks

- Old: clip.std() > 0.01 accepts a frozen title card and low-motion clips
- New gate rejects frozen, black, blurry, low-motion (abnormal classes), clips spanning a cut, fades and flashes
- Blur floor from S1: half of the 5th percentile of sharpness
- Original motion-fingerprint dedup (0.99) kept unchanged

S9: packaged clips and a dataset report



Per-class motion, entropy, people, brightness, gate rejections



One saved clip (8 of 16 frames)

2.4 MB

per clip as uint8, versus 9.6 MB as float32

- JSON sidecar per clip: shot id, motion stats, people, crop box, brightness, enhanced?
- Same folder layout as before

What changed, measured

Operation	Before	After
Clip frame rate	15 FPS (step 2)	**12.00 FPS (timestamps)**
Framing	Stretched 640 × 640	**Letterbox + actor-centred crop**
Brightness rule	Three different checks	**One luma measure, threshold 50**
Clip sampling	Uniform, label noise	**Inside proposals, hard negatives**
Validity check	std > 0.01	**Six-rule quality gate**
Tracking	None committed	**ByteTrack + interaction signals**
Clip size	9.6 MB (float32)	**2.4 MB (uint8) + sidecar**

Limitations and next steps

Limitations

- Activity thresholds tuned on three annotated videos
- S5 skipped: two moving-camera videos remain
- Overlay detector over-fires on textured borders
- No ground truth for person detection on UCF-Crime

Next steps

- Full extraction: about 3 hours, 24 GB for four folds
- Retrain VideoMAE on the new clips (loader needs /255)
- Ablations: sampling, framing, enhancement, cleaning
- Use the shared brightness rule in the backend

Better processing before training

1. Five silent preprocessing gaps found and fixed, each with a measurable input and output sample.
2. Video analytics (motion, tracks, activity) now decides which clips the model sees.
3. Honest findings: the fine-tuned detector does not transfer; proposals are recall-heavy.

Questions?